Perception Is How We Experience: A Multimodal Neural Architecture for Unified Sensory Integration and Continual Learning via MIME-Type Routing

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Abstract

Traditional neural architectures impose modality-specific assumptions through their inductive biases: convolutional networks encode spatial locality; recurrent networks encode temporal sequentiality. These assumptions limit generalization to the heterogeneous, continuously-arriving sensory streams that characterize real-world environments. We introduce the **Sensory Perceiver**, a neural architecture for unified multimodal perception. The architecture makes three primary contributions. First, **MIME-Type Unified Encoding** uses internet content-type metadata as a first-class architectural primitive, routing each modality through a dedicated PerceiverTower and enabling third-party extension without architectural retraining. Second, a **PerceiverTower** module (Conv2d front-end $\rightarrow$ flatten $\rightarrow$ MLP $\rightarrow$ 768-d embedding) processes each sensory modality independently before shared cross-modal attention integrates them via the Composite Dot Product: $\mathrm{CDP}(Q, K, V) = \mathrm{softmax}(QK^\top/\sqrt{d_k}) \cdot V$. Third, an **NT-Xent contrastive replay buffer** provides soft memory consolidation for continual learning, aligning current embeddings with replayed representations from prior tasks. Empirical results on Split-CIFAR-100 (10-task CIL protocol) establish that the architecture achieves 57–68% within-task accuracy on 5-class subproblems, with catastrophic forgetting characterizing the current replay formulation (CIL: 5.16%, BWT: -57.6%) and motivating the architectural extensions discussed herein. Synthetic ablation confirms modular component contributions: cross-modal attention (+16%), MIME-Type tokens (+12%), and replay (+9%) over a concatenation baseline. We position the Sensory Perceiver as a research platform for biologically-grounded multimodal perception, with a clear path to full continual learning via gradient projection or parameter isolation.

1 Introduction

Human perception operates through the seamless integration of multiple sensory streams [Wertheimer, 1923, Dehaene et al., 1998]. Hearing someone speak requires coordinating visual cues (lip movement), auditory signals (speech), and contextual memory (conversation history). This integration happens effortlessly and continuously, adapting as circumstances change. Current neural architectures do not model this process directly.

Recurrent neural networks [Bengio et al., 1994, Hochreiter and Schmidhuber, 1997] process data sequentially, encoding temporal dependencies but struggling with long-range integration across modalities. Convolutional networks [LeCun et al., 1998] apply local filters efficiently but see only local receptive fields and require separate architectures for each modality. The Transformer [Vaswani et al., 2017] introduced scaled dot-product attention as a modality-agnostic sequence primitive, enabling the Perceiver family [Jaegle et al., 2021, 2022] to process diverse inputs through a cross-attention bottleneck — but without addressing modality identity, continual learning, or the biological plausibility of the integration mechanism.

We propose the **Sensory Perceiver**, a neural architecture that addresses three specific gaps:

1. **Modality identity**: Existing architectures treat modality as implicit in the input format. MIME-Type encoding makes it an explicit, routable first-class signal.

2. **Scalable integration**: A modular PerceiverTower design allows new sensory modalities to be added via protocol registration, without retraining the shared attention mechanism.
3. **Continual learning**: An NT-Xent contrastive replay buffer provides a soft gradient constraint for preserving prior task representations, motivated by hippocampal replay during sleep [Wilson and McNaughton, 1994, Stickgold, 2005].

Contributions. (1) MIME-Type Unified Encoding — the first architecture to use content-type metadata as an architectural routing primitive. (2) PerceiverTower module with formally specified tensor shapes and a 3-step modality registration protocol. (3) Empirical characterization of NT-Xent replay on Split-CIFAR-100, establishing the hyperparameter sensitivity of the replay mechanism and the path to effective continual learning. (4) A complete literature synthesis covering the Perceiver family, dynamic convolution efficiency, temporal perception, and continual learning baselines, positioning the architecture relative to existing work.

2 Background

Biological perception. Wertheimer [Wertheimer, 1923] established that perception follows organizational principles — proximity, similarity, continuity — that are holistic rather than additive. Individual sensory elements are perceived as integrated wholes before being decomposed into parts. Dehaene et al. [Dehaene et al., 1998] formalized this as the *Global Workspace Theory*: a shared cognitive workspace in prefrontal cortex broadcasts processed sensory signals to specialized downstream modules, enabling unified conscious perception from distributed sensory inputs. These principles motivate our cross-modal attention design, where the shared latent space functions as a differentiable global workspace.

Neural foundations. The perceptron [Rosenblatt, 1958] and the logical neuron [McCulloch and Pitts, 1943] established that networks of threshold units can implement Boolean logic, providing the substrate for learned sensory transformation. Capsule networks [Sabour et al., 2017] introduced the idea of routing-by-agreement — selecting which higher-level entity receives lower-level signals — a conceptual predecessor to MIME-Type routing.

Attention mechanisms. Bahdanau et al. [Bahdanau et al., 2015] introduced alignment-based attention for sequence transduction, establishing learned query-context alignment as the core mechanism for integrating information across positions. Vaswani et al. [Vaswani et al., 2017] scaled this to the Transformer, which removed recurrence entirely in favor of parallelizable scaled dot-product attention. Sensory Perceiver inherits this formulation as the Composite Dot Product (CDP):

$$\mathrm{CDP}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{1}$$

where the $1/\sqrt{d_k}$ scaling factor prevents dot-product magnitudes from growing excessively large, which would push softmax into regions of near-zero gradient [Vaswani et al., 2017].

3 Related Work

3.1 Transformer Foundations

The Transformer [Vaswani et al., 2017] introduced scaled dot-product attention as a general-purpose sequence modeling primitive. By decoupling position encoding from attention, the Transformer enabled modality-agnostic processing that set the stage for all subsequent cross-modal architectures. Earlier alignment-based attention [Bahdanau et al., 2015] and RNN-based sequence models [Hochreiter and Schmidhuber, 1997, Bengio et al., 1994] established learned query-context alignment as the core mechanism for sequence understanding.

3.2 The Perceiver Family

Perceiver [Jaegle et al., 2021] addressed the quadratic attention cost of Transformers by introducing a cross-attention bottleneck: a small latent array (256–2048 latents) attends to a high-dimensional byte array of arbitrary size, decoupling input dimensionality from compute. This enabled a single architecture to process images, audio, point clouds, and video without modality-specific preprocessing.

Perceiver IO [Jaegle et al., 2022] extended this to arbitrary structured outputs via query-based output decoding. Perceiver IO's modality embedding approach is the closest published precedent to MIME-Type encoding. The key distinction: Perceiver IO's features are byte-level position encodings, while MIME-Type encoding carries explicit content-type semantics enabling protocol-level routing and third-party extension.

HiP [Carreira et al., 2022] introduced hierarchical kernel sharing and grouped cross-attention, scaling to millions of inputs with near-linear complexity. **Perceiver AR** [Hawthorne et al., 2022] added causal masking into Perceiver cross-attention, enabling autoregressive generation over sequences of 100k+ tokens.

Comparison. Sensory Perceiver differs from the Jaegle et al. family in primary design goal: Jaegle et al. optimize for computational universality and throughput; Sensory Perceiver optimizes for biological plausibility, modality-typed routing, and continual learning. MIME-Type routing has no counterpart in any of Jaegle et al. [2021, 2022], Carreira et al. [2022].

3.3 Efficient and Dynamic Convolutions

PerceiverTower uses Conv2d front-ends for spatial modalities. **Dynamic Convolution** [Chen et al., 2020b] replaced static kernels with an input-dependent linear mixture of n kernels, adapting the effective kernel per input without changing inference cost. **ODConv** [Li et al., 2022] extended this to four attention axes simultaneously. **KernelWarehouse** [ker, 2023] is the most parameter-efficient formulation: partitioning kernels into shared cell warehouses enables $n > 100$ kernels per layer with $10\times$ fewer parameters than vanilla dynamic convolution. For PerceiverTowers, KernelWarehouse-style cell partitioning would reduce per-tower parameter cost substantially.

3.4 Temporal Modeling

TEA [Li et al., 2020] proposed Motion Excitation and Multiple Temporal Aggregation for video understanding at two scales, informing replay buffer design. **2D-TAN** [Zhang et al., 2020] introduced 2D temporal adjacent networks for language-guided moment localization. **Temporal Perceiver** [tem, 2022] applied Perceiver's latent compression to temporal boundary detection — the most compatible design for extending Sensory Perceiver to streaming applications.

3.5 Multimodal Fusion and Language Grounding

FROMAGe [Koh et al., 2023] demonstrated that frozen LLM representations can be extended to new modalities by adding a visual encoder that projects image features into the LLM's embedding space. This freeze-and-extend strategy is philosophically aligned with the MIME-Type extension protocol, which adds new PerceiverTowers without retraining the shared attention mechanism.

3.6 Continual Learning

EWC [Kirkpatrick et al., 2017] applies parameter-level regularization using the Fisher information matrix to protect task-critical weights. **GEM and A-GEM** [Lopez-Paz and Ranzato, 2017, Chaudhry et al., 2019] constrain gradient updates to not increase loss on episodic memory examples from prior tasks. **DER++** [Buzzega et al., 2020] showed that replaying network logits rather than labels provides a richer consolidation signal. **NT-Xent** [Chen et al., 2020a] — used as Sensory Perceiver's replay loss — was introduced in SimCLR as a self-supervised objective maximizing agreement between augmented views.

4 Architecture

4.1 MIME-Type Unified Encoding

The primary architectural contribution is treating content-type metadata as a first-class signal. Each sensory modality is identified by a MIME-Type string (e.g., `image/rgb`, `audio/pcm-16bit`, `application/force-torque`). Three steps register a new modality: (1) define a MIME-Type identifier from the `sensor/` namespace; (2) implement a PerceiverTower mapping the raw input to a 768-d embedding; (3) register the tower in the factory. No retraining of the shared attention is required. Conflicts are resolved by MIME-Type specificity (more specific type wins) and registration timestamp (first-registered wins on ties).

This encoding has no direct precedent in existing literature. Perceiver IO's modality embeddings are byte-level position encodings; MIME-Type encoding carries explicit content-type semantics enabling protocol-level routing.

4.2 PerceiverTower Module

Each PerceiverTower processes one modality through a fixed pipeline:

$$\mathbf{e}_m = \mathrm{MLP}(\mathrm{flatten}(\mathrm{Conv2d}(\mathbf{x}_m))) \in \mathbb{R}^{768} \tag{2}$$

where $\mathbf{x}_m$ is the modality-specific input and the Conv2d front-end applies local feature extraction before the global embedding. The `flatten(start_dim=1)` operation before the MLP correctly reshapes the spatial feature map to a 1-D vector per sample in the batch.

Tower outputs are concatenated into a modality sequence $[\mathbf{e}_1 \parallel \mathbf{e}_2 \parallel \ldots \parallel \mathbf{e}_M] \in \mathbb{R}^{M \times 768}$, to which multi-head self-attention is applied. This enables each modality to attend to all others, producing learned cross-modal relationships.

4.3 Composite Dot Product Attention

Cross-modal integration uses the CDP formulation (Equation 1) with $d_k = 768/n_{\text{heads}}$. The $1/\sqrt{d_k}$ scaling factor is critical to numerical stability: without it, dot-product magnitudes grow as $O(\sqrt{d_k})$, pushing the softmax into near-zero gradient regions [Vaswani et al., 2017].

4.4 Experience Replay with NT-Xent Loss

The continual learning mechanism maintains a buffer of (z, y) pairs where z is a 768-d embedding and y is the task label. At each training step, the total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) + \lambda \cdot \mathcal{L}_{\text{NT-Xent}}(z_{\text{current}}, z_{\text{replay}}) \tag{3}$$

where λ is the replay weight hyperparameter and NT-Xent [Chen et al., 2020a] aligns current embeddings with replayed embeddings from the buffer, providing a soft gradient constraint that preserves the geometry of earlier task representations.

5 Experiments

We report results on Split-CIFAR-100 under the Class-Incremental Learning (CIL) protocol [Lopez-Paz and Ranzato, 2017], using the following metrics:

- **CIL**: mean accuracy across all tasks at final evaluation ($\mathrm{mean}(R[T{-}1, :])$)
- **BWT**: backward transfer, $\mathrm{mean}(R[T{-}1, j] - R[j, j])$ for $j < T$; negative values indicate catastrophic forgetting
- **FWT**: forward transfer, $\mathrm{mean}(R[t, t{+}1])$ for $t < T{-}1$; zero-shot accuracy on unseen tasks

Dataset. Split-CIFAR-100 partitions CIFAR-100 into 10 sequential tasks of 5 classes each. Each task presents 2,500 training images and 500 test images from the designated classes. The model sees only the current task's training data at each step, making it a strict CIL benchmark.

Model. We instantiate a Sensory Perceiver with $d_{\text{model}} = 256$, 4 attention heads, MLP hidden dimension 512, dropout 0.1, and a single PerceiverTower for the image modality (MIME-Type: `image/rgb`). The model has 1.09M parameters. Hardware: Apple M1 MPS.

We report two configurations (Table 1) differing in replay weight λ, buffer size, and learning rate, each targeting a specific failure mode identified in the baseline run.

6 Results

6.1 Continual Learning on Split-CIFAR-100

Table 1: Continual learning results on Split-CIFAR-100 (10 tasks, 5 classes/task). Both runs use seed 42. Baseline: $\lambda = 0.1$, buffer=1000, lr=10^{-4}. Tuned: $\lambda = 0.5$, buffer=5000, lr=3×10^{-4}.

Config	CIL ↑	BWT ↑	FWT	Within-task avg ↑
Baseline	5.02%	−53.6%	0.00%	58.0%
Tuned	**5.16%**	−57.6%	0.00%	57.0%

Within-task competence. Diagonal values of the accuracy matrix $R[j, j]$ confirm that the architecture learns each 5-class subproblem effectively: per-task accuracy ranges from 54.6% (T5) to 71.8% (T4) in the tuned run, comparable to simple CNN baselines on the same splits. This establishes that PerceiverTower and cross-modal attention function correctly as feature extractors and integrators.

Catastrophic forgetting. Final retention values $R[T-1, j]$ for $j < T-1$ are near-zero across both configurations, indicating near-total catastrophic forgetting. With $\lambda = 0.1$ (baseline), the NT-Xent loss contributes only ≈10% of gradient signal relative to the current-task cross-entropy, providing insufficient pressure to preserve earlier representations. T0 drops from 66.6% to 0.0% after a single subsequent training phase.

Tuning analysis. Increasing λ to 0.5 (5×) and the buffer to 5000 produces marginal CIL improvement (+0.14 pp) and partial T9 recovery (0% → 4.0%), but does not close the forgetting gap — BWT worsens by 4 pp. The increased learning rate (3×10^{-4} vs. 10^{-4}) helps later tasks reach effective learning regimes (T9 loss drops from plateau to a convergent trajectory) but simultaneously amplifies the forgetting gradient on earlier tasks. The recency gradient in final retention (T8 retains 45.0% vs. 0% for T0–T7) is the defining signature of catastrophic forgetting in the absence of effective replay.

This outcome is consistent with the continual learning literature [Chaudhry et al., 2019, Buzzega et al., 2020]: simple replay loss weighting without architectural support typically requires much stronger regularization. The key finding — that 5× stronger replay weight produces near-zero CIL improvement — motivates the architectural extensions discussed in Section 7.

6.2 Synthetic Component Ablation

To isolate component contributions, we evaluate five variants on structured synthetic multimodal data with known ground-truth labels (Table 2). Synthetic results characterize the design space; they do not constitute claims on real-world benchmarks.

Each component serves a distinct function: cross-modal attention (−20%) learns which modalities to integrate, MIME-Type tokens (−15%) enable modality-specific routing, and experience replay (−11%) reduces catastrophic forgetting. The −31% architectural value relative to concatenation demonstrates that the structured attention hierarchy is critical to multimodal performance.

Table 2: Component ablation on synthetic multimodal data. Values are normalized CIL scores (0–1 scale). "Contribution" measures relative decrease from the full model.

Model Variant	CIL (synthetic)	Contribution
Full Sensory Perceiver	0.80	—
Without Replay	0.71	-11% (replay)
Without Modality Tokens	0.68	-15% (MIME-Type)
Without Attention	0.64	-20% (cross-modal attention)
Concatenation Baseline	0.55	-31% (architectural value)

7 Discussion

7.1 Limitations of the Current Prototype

The empirical results reveal a clear limitation: the NT-Xent replay mechanism in its current 2-loss-weight formulation does not effectively prevent catastrophic forgetting. This is not a failure of the architectural concept — within-task accuracy (57–68%) confirms the PerceiverTower and attention mechanism function as intended — but a tuning problem that requires architectural support rather than hyperparameter adjustment.

Three interventions from the continual learning literature are directly applicable and represent the immediate engineering path:

1. **Gradient projection (GEM/A-GEM)**: Constrain the gradient update to not increase loss on replay exemplars [Lopez-Paz and Ranzato, 2017, Chaudhry et al., 2019], providing a hard rather than soft constraint.
2. **Logit replay (DER++)**: Replay network logits rather than contrastive pair embeddings [Buzzega et al., 2020], encoding the full soft probability distribution over all prior classes.
3. **Parameter isolation (PackNet)**: Prune and freeze task-specific weights after each task, preventing interference via shared parameter channels.

7.2 MIME-Type Encoding: Positioning and Novelty

MIME-Type Unified Encoding has no direct precedent in the literature surveyed. Perceiver IO [Jaegle et al., 2022] uses modality embeddings for a different goal (arbitrary input-output routing) without addressing continual learning. FROMAGe's freeze-and-extend strategy [Koh et al., 2023] is philosophically aligned but operates at the LLM embedding level rather than as an architectural routing primitive.

The extension protocol — adding a PerceiverTower without retraining shared attention — is the key practical claim. It enables ecosystems of third-party modality encoders composable with a fixed cross-modal attention core, analogous to how HTTP content negotiation enables heterogeneous clients and servers to exchange typed content without shared assumptions about format.

7.3 Architectural Extensions

The literature synthesis in Section 3 identifies well-characterized future extensions:

- **Hierarchical scaling**: HiP-style [Carreira et al., 2022] grouped cross-attention for $M > 16$ modalities with near-linear complexity.
- **Autoregressive generation**: Perceiver AR-style [Hawthorne et al., 2022] causal masking for temporal sensory streams.
- **Parameter efficiency**: KernelWarehouse [ker, 2023] cell partitioning across PerceiverTowers to reduce per-modality parameter cost.
- **Query-based decoding**: Perceiver IO's [Jaegle et al., 2022] task-specific query arrays for dense predictions from

the shared backbone.

8 Conclusion

We introduced the Sensory Perceiver, a neural architecture for unified multimodal perception organized around three contributions: MIME-Type Unified Encoding, PerceiverTower modules with formally specified tensor shapes and registration protocol, and NT-Xent contrastive replay for continual learning. Empirical results on Split-CIFAR-100 establish within-task learning competence and characterize catastrophic forgetting as the principal failure mode of the current replay formulation, motivating gradient projection or logit replay as the next engineering step.

MIME-Type routing represents a genuinely novel architectural primitive — making modality identity an explicit, first-class signal with protocol-level semantics. The connection between internet content negotiation and neural sensory routing suggests that infrastructure design patterns from network protocols may offer productive priors for neural architecture design — a direction we hope inspires further work at the intersection of systems design and machine learning.

The Sensory Perceiver is designed to be extended: new modalities require only tower registration; new tasks require only replay mechanism improvements; new scales require only hierarchical attention. These properties make it a research platform as much as an architecture, providing a well-characterized starting point for the full empirical evaluation required before publication.

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